

Furthose samreen B(192511164)

CO1-AT2 ASSESSMENT TOOL

1. Backtracking Search (Hospital Doctor Assignment)

Given

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Constraints

D1 $\neq$ Night

D2 works before D3 ($M < A < N$)

Only one doctor per shift

D3 $\neq$ Morning

Every doctor gets one shift

Backtracking Steps

Step

Assignment

Constraint Check

Result

1

D1 $\rightarrow$ Morning

✓ Allowed

Continue

2

D2 $\rightarrow$ Afternoon

✓ Before D3

Continue

3

D3 → Night

✓ Not Morning, D2 before D3

Valid

No backtracking is needed because the first assignment satisfies all constraints.

Final Schedule

Doctor

Shift

D1

Morning

D2

Afternoon

D3

Night

2. BFS Search (Robot Grid 5×5)

Assume Grid:

S . . X .

. X . X .

. X . . .

. . X X .

. . . G .

S = Start, G = Goal, X = Obstacle

Node Expansion

Start → (0,0)

Expand → (0,1), (1,0)

Expand → (0,2)

Expand $\rightarrow (1,2)$

Expand $\rightarrow (2,2)$

Expand $\rightarrow (2,3)$

Expand $\rightarrow (2,4)$

Expand $\rightarrow (3,4)$

Expand $\rightarrow (4,4)$

Expand $\rightarrow$ Goal $(4,3)$

Queue Updates

$Q = [S]$

$Q = [(0,1),(1,0)]$

$Q = [(1,0),(0,2)]$

$Q = [(0,2),(1,2)]$

$Q = [(1,2),(2,2)]$

$Q = [(2,2),(2,3)]$

$Q = [(2,3),(2,4)]$

$Q = [(2,4),(3,4)]$

$Q = [(3,4),(4,4)]$

Goal reached

Optimal Path

$S \rightarrow \text{Right} \rightarrow \text{Right} \rightarrow \text{Down} \rightarrow \text{Down} \rightarrow \text{Right} \rightarrow \text{Right} \rightarrow \text{Down} \rightarrow \text{Down} \rightarrow \text{Left} \rightarrow G$

Total Cost

$10 \text{ moves} \times 1 = 10$

3. Autonomous Rescue Robot

a) Constraint Analysis

Moves only Up, Down, Left, Right.

Obstacles cannot be crossed.

Robot senses only adjacent cells.

Normal move cost = 1.

Risky cell cost = 3 (1 + 2).

Robot updates map while moving (Online Search).

b) Solution (Uniform Cost Search + Online Search)

Start from S.

Sense nearby cells.

Add safe paths to the priority queue.

Choose the path with the lowest cost.

Update the map if new obstacles are found.

Continue until reaching G.

c) AI Concepts

Intelligent Agent

Perceives surroundings using sensors.

Decides the safest and least-cost path.

Acts using movements.

Rationality

Chooses the minimum-cost safe path.

Avoids risky and blocked cells.

Environment Type

Partially observable.

Dynamic.

Sequential.

Discrete.

d) Justification

Uniform Cost Search always finds the minimum-cost path, even when risky cells have extra cost.

Online Search is suitable because the robot discovers obstacles while moving.

Together they provide safe, optimal, and adaptive navigation in a disaster environment.

Conclusion: Uniform Cost Search with an Online Search Agent is the best approach because it minimizes cost, avoids obstacles, and updates decisions dynamically.